

APS110 – Lecture 34

See hand written notes for drawings.

Demo → salt onto ice

- When salt is sprinkled onto ice, the ice will melt
- What is happening thermodynamically?
 - Melting is endothermic
 - Recall ice formation (freezing of water) is exothermic
 - So we should expect the temperature to DECREASE
 - With just a little bit of salt, the ice drops down to -27°C almost instantly
- Next, ice cream maker with Dr. Pepper
 - Sprinkle salt onto ice surrounding ice cream container → temperature rapidly drops
 - Enthalpy of fusion is bringing the temperature of the ice down
 - Dr. Pepper has freezing point depression because of the high sugar content
 - Will the Dr. Pepper freeze solid?
 - Will form slush → deals with phase equilibria (to be discussed later)

Recall: ΔH vs. ΔU (change in internal energy)

- Ex. combustion of gasoline
 - Gasoline can be approximated as octane
 - $2 \text{C}_8\text{H}_{18(l)} + 25 \text{O}_{2(g)} \rightarrow 18 \text{H}_2\text{O}_{(g)} + 16 \text{CO}_{2(g)}$
 - Molecules of CO_2 are released in this rxn → what is the tax on the atmosphere?
 - Calculate the enthalpy of combustion:
 - *** Important to state the temperature for the rxn
 - We typically state STANDARD CONDITIONS → denoted by °
 - 1 bar & 298.15K

$$\Delta_{\text{combustion}} H^\circ = 16 \Delta_f H^\circ \text{CO}_{2(g)} + 18 \Delta_f H^\circ \text{H}_2\text{O}_{(g)} - 2 \Delta_f H^\circ \text{C}_8\text{H}_{18(l)} - 25 \Delta_f H^\circ \text{O}_{2(g)}$$

$$\Delta_{\text{combustion}} H^\circ = 16 (-394 \text{ kJ/mol}) + 18 (-242) - 2 (-250) = -10,160 \text{ kJ per } \underline{2 \text{ mol}} \text{ of octane}$$

- Or, -5080 kJ per mol of C_8H_{18}
- What is the enthalpy required to make a mole of water vapour? What is the enthalpy required to form CO_2 gas?
 - These are all **standard formation energies**, that have been calculated and tabulated
- We need to use NEGATIVE $\Delta_f H^\circ$ for when the molecule is breaking (reagents)
- **Because oxygen is in its stable form already (O_2 in gas phase), the formation energy is ZERO***
 - Already in standard condition, so no energy to get O_2 into its standard condition (already in standard condition)
- What about PV → $P\Delta V$ work?

- 34 mol created
 - 25 mol consumed
 - Total 9 mol created
- So, $W = P\Delta V = \Delta(nRT)$
 - $W = 9 \times 8.314 \times 298 = 22 \text{ kJ}$
 - Since $\Delta H = \Delta U + P\Delta V \rightarrow \Delta U = \Delta H - P\Delta V = -10,160 - 22 = -10,182 \text{ kJ}$
- **Standard state** $\rightarrow$ what we use to measure everything against
 - Enthalpy changes are measured relative to this standard state
 - **Defined as MOST STABLE form of the element at standard pressure (1 bar) and temperature (298.15K)**
- Enthalpy is a **STATE VARIABLE**
 - Only start and end points matter, we don't care about how we get from start to end
 - Potential energy is also a state variable
 - Other state variables:
 - Pressure (P)
 - Entropy (S)
 - Temperature (T)
 - Internal Energy (U)
 - Volume (V)
 - Because of this, **we can ADD and SUBTRACT state variables**
 - Can add and subtract to get the reaction enthalpy (like previous example with octane)
- Heat (q) & work (W) are **PATH VARIABLES**
- Entropy of a pure crystal approaches ZERO ENTROPY as temperature approaches 0 K
 - Recall all electrons occupy lowest energy state
 - It's because of this that we can measure absolute values of entropy!
 - Entropy is calculated in absolute terms
- Consider $\text{H}_2\text{O}_{(s)} \rightarrow \text{H}_2\text{O}_{(l)}$
 - Phase transformation
 - Graph (T vs. heat supplied)
 - Two plateaus $\rightarrow$ correspond to the heat required for phase transformation
 - Slopes represent the HEAT CAPACITY of the substance
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